

Folding Method in Hashing

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Folding Method in Hashing: It breaks up a key value into precise segments that are added to form a [hash value](#), and look at another technique is to apply a multiplicative hash function to each segment individually before adding. Some folding methods go one step further and reverse every other piece before the addition. This folding method is **independent of distribution**.

Algorithm:

- The folding method is used for creating hash functions starts with the item being **divided into equal-sized pieces i.e.**, the last piece may not be of equal size.
- The outcome of adding these [bits](#) together is the hash value, $H(x) = (a + b + c) \bmod M$, where a, b, and c represent the preconditioned key broken down into three parts and **M is the table size, a** and **mod** stands for [modulo](#).
- In other words, the sum of three parts of the preconditioned key is divided by the table size. The remainder is the hash key.

Explanation:

Example 1: The task is to fold the key **123456789** into a Hash Table of ten spaces (0 through 9).

- It is given that the key, say **X** is 123456789 and the table size (i.e., **M** = 10).
- Since it can break **X** into three parts in any order. Let's divide it evenly.
- Therefore, a = 123, b = 456, c = 789.
- Now, $H(x) = (a + b + c) \bmod M$ i.e., $H(123456789) = (123 + 456 + 789) \bmod 10 = 1368 \bmod 10 = 8$.
- Hence, 123456789 is inserted into the table at address **8**.

Example 2: The task is to fold the key **452378912** into a Hash Table of ten spaces (0 through 9).

- It is given that the key, say **X** is **452378912** and the table size (i.e., **M** = 10).
- Since it can break **X** into three parts in any order. Let's divide it evenly.
- Therefore, a = 452, b = 378, c = 912.

- Now, $H(x) = (a + b + c) \bmod M$ i.e., $H(452378912) = (452 + 378 + 912) \bmod 10 = 1742 \bmod 10 = 2$.
- Hence, 452378912 is inserted into the table at address 2.

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